

TOPIC 3.4

How does energy transformation affect humans?

Key Concepts

- Energy transformations provide benefits, but not without risk of harm.
- We take action and create technology to reduce the harmful impacts of energy transformation.
- We learn from our mistakes and each other to reduce harmful impacts of energy transformation.

Curricular Competencies

- Demonstrate an awareness of assumptions and bias.
- Consider social, ethical, and environmental implications of findings from investigations.
- Contribute to care for self, others, community, and world through individual or collaborative approaches.

Imagine a world where you find yourself wading across roads, or even your own front yard. It's a reality that the residents of the South Pacific island nation of Tuvalu face at every high tide. Increasing global temperatures are melting polar ice caps and glaciers around the world. Melt water from these processes is already flooding some low-lying coastlines and islands, like Tuvalu. Although not all islands are at risk, some face a future of being lost beneath the waves. Many coastal cities in BC will also be affected by rising sea levels. The Lower Mainland is projected to see a 30 cm increase in sea level by 2050. It won't lead to a Tuvalu-type scenario just yet. However, such an increase may increase flooding of low-lying coastal neighbourhoods and erosion of beaches.



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Considering** Why are warmer global temperatures leading to a rise in sea level? (Hint: There is more than one reason.)
2. **Applying** How might a rise in sea level affect you if you live on the coast? What if you do not live on the coast? Discuss your ideas.
3. **Applying First Peoples Perspectives** How can the ideas of renewal and reciprocity be applied to guide the ways that we use energy transformations for human benefit now and for future generations?



CONCEPT 1

Energy transformations provide benefits, but not without risk of harm.

Connect to Investigation 3-H on page 274

All living things rely on energy transformations, and we are no exception. Locally and globally, human societies develop technologies and understandings that enable us to use and benefit from energy transformations. These benefits, however, rarely come without the risk of negative effects. The time line below provides highlights of our use of energy transformations.

1 500 000 B.C.E.

Early humans relied on plants' abilities to transform solar energy into chemical potential energy through photosynthesis. Through cellular respiration, they transformed this energy into various other forms of energy to carry out life functions.



200 B.C.E.–200 C.E.

Wood was now burned on a large scale. For example, the ancient Roman Empire and the Chinese Han dynasty burned great amounts of wood to form metal for weapons and to heat their homes and baths. However, this also caused the first environmental health problems in recorded history.



1698–1900 C.E.

Combustion of coal in steam engines transformed chemical potential energy to thermal energy to mechanical kinetic energy. The kinetic energy was transferred to the vehicle or machinery run by the steam engine. However, burning coal also increased air pollution. This was especially harmful in London, England. Several times in history, coal smoke mingled with fog, covering the city in a deadly blanket. Many lives were lost to bronchitis, pneumonia, and even heart failure.



1 500 000 B.C.E. 1 000 000 500 000 0 C.E. 1600 1700

1 000 000–500 000 B.C.E.

Fire-starting technology let early humans transform chemical potential energy in firewood into thermal and radiant energy. Food could be cooked, and the night could now be lit up. Also, people could survive in much colder climates.



125 C.E.

Cement was used by Romans to build architectural wonders, such as the roof of the Pantheon shown here. Today we produce more than 4.2 trillion tonnes of cement each year. The chemical reaction that produces cement transforms chemical potential energy to thermal energy. It also adds 5% of global carbon dioxide emissions to the atmosphere.



1800 C.E.

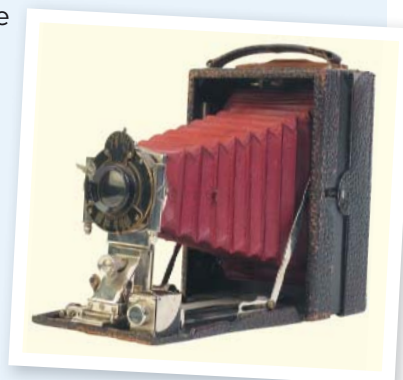
The first working battery was invented by Italian physicist Alessandro Volta. Batteries involve the transformation of chemical potential energy to electrical potential energy. Modern batteries have become a landfill problem. Also, those that are not recycled properly leach toxic chemicals that harm human health.



Activity

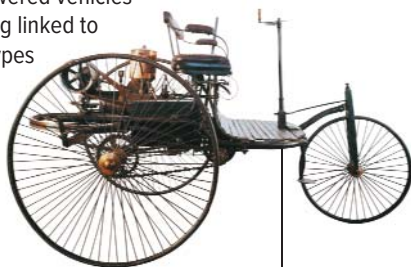
Snapshots of History

1. Through discussion, research, or materials provided by your teacher, find three ways we use energy transformations that are not on the time line.
2. For each example, record the dates it occurred. Then, write a short summary that captures a snapshot of that moment in history. Note any benefits or harm caused by the energy transformation.
3. Use a camera to take a photographic snapshot of an object or scene that visually represents your historical snapshot.
4. Add your snapshots, written and photographic, to your own class time line of human energy transformation. Your teacher may ask you to present your snapshots to the class.



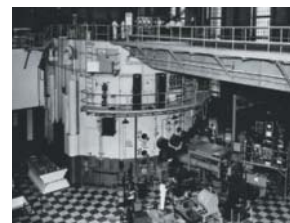
1859 C.E.

The first continuously operating gas engine was built by French engineer Étienne Lenoir. This transformation of chemical potential to mechanical kinetic energy paved the way for the automobile. However, by the 1950s, cars and other gas-powered vehicles were already being linked to smog and other types of air pollution.



1945 C.E.

The Canadian ZEEP nuclear reactor was one of the world's first nuclear reactors. The reactor transformed nuclear energy into thermal energy, which was used to make steam. The energy of the steam was transferred to a generator, which transformed it into electrical energy. Additionally, radioactive isotopes created in nuclear reactors are used to treat and diagnose a variety of medical conditions. Despite these benefits, the fact remains that nuclear radiation is harmful to all life, including humans.



1800

1900

2000

1882 C.E.

The first hydroelectric dam was built to transform gravitational potential and mechanical kinetic energy into electrical energy. However, dam construction can flood vast areas of land. This releases large amounts of the greenhouse gas methane into the atmosphere as plants decay. Also, flooding may displace human populations, often First Peoples, living in the area.



1959 C.E.

Harry Ihrig designed the first fuel cell to run a vehicle. A few years later, the technology was used to provide electrical power to the Gemini and Apollo spacecraft. However, producing the hydrogen used in fuel cells adds carbon dioxide to the atmosphere.



Before you leave this page . . .

1. What do you think is the greatest impact, positive or negative, of our application of energy transformations. Justify your response.

CONCEPT 2

We take action and create technology to reduce the harmful impacts of energy transformation.

Reducing the impacts of technology-related energy transformation requires personal and national action. This often involves making choices, such as reducing energy use and choosing sustainable technologies to lessen harmful impacts. However, taking action requires commitment and sometimes making sacrifices in comfort and convenience. Despite these challenges, we can and are taking action to reduce these harmful effects. **Figure 3.39** shows some of the ways students like you are making a difference.

Figure 3.39 Student initiatives that reduce energy use and support sustainable energy-transformation technologies can bring about change.

Applying: What initiative could you get involved in?

Many students cycle to school rather than travel by car. This also reduces combustion-related pollution.



Urban farm programs like the one at Victoria High School on Vancouver Island help reduce pollution caused by transporting food over long distances.

Through participating in recycling programs, students in northern BC and Yukon have kept thousands of phonebooks out of landfills. This has helped reduce harmful impacts of manufacturing-related energy transformation.



The text and visuals below explore emerging technologies that can help address problems such as air pollution and climate change. Some people believe that such technologies will be a cure-all for the harmful effects of energy transformation. Yet, technology is rarely problem free. For instance, electric vehicles are emission free. However, generating the electrical energy they use may not be. As well, their batteries contain toxic metals that can leach into the landfill if not recycled properly.

Methane Cracking

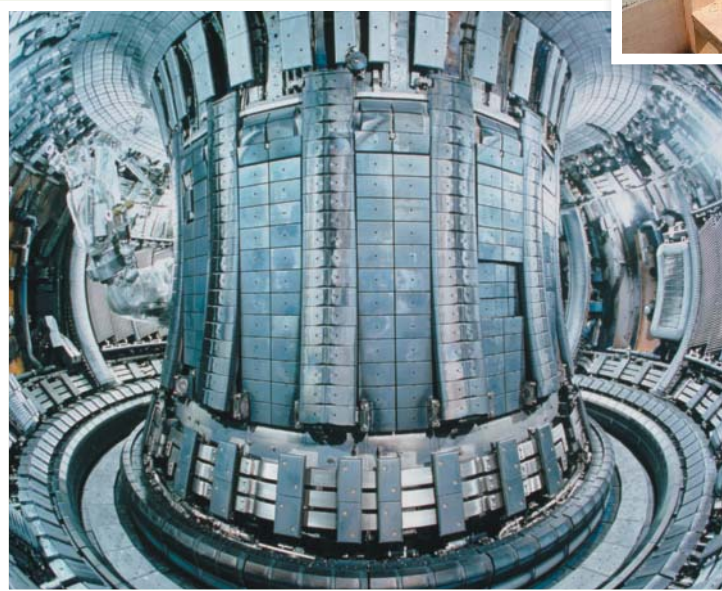
Hydrogen fuel cells are a good example of a type of “clean” energy with hidden harmful impacts. The fuel cells emit no pollutants. But the process that is normally used to produce hydrogen, called steam methane reforming, still releases about half the carbon dioxide of a gas-powered vehicle. An emerging technology known as *methane cracking* can now produce hydrogen from methane without releasing carbon dioxide. Some carbon dioxide is still released, but emissions are very low compared to steam methane reforming.



Cement Production

The chemical reaction used to create lime for cement production releases carbon dioxide. Since 2008, work has been underway to develop a way of making cement that produces about 70 percent fewer carbon dioxide emissions. Considering how much cement is made worldwide, this

invention could significantly reduce greenhouse gas emissions. Also, researchers are trying to create cement that actually absorbs carbon dioxide as it hardens.



Nuclear Fusion

For more than 50 years, scientists have been searching for a way to control nuclear fusion to generate electrical energy. A very high temperature is needed to start a fusion reaction. But scientists have already created temperatures hotter than the Sun. The real challenges are to sustain these temperatures and to contain the super-heated reaction. No known material can withstand such scorching temperatures. Yet a solution may be in the works. A promising technology, called a tokamak reactor, uses powerful magnetic fields to contain the reaction, and keep it away from the reactor walls. In 2017, a tokamak reactor in China was able to contain fusion at a temperature nearly three times that of the Sun. It contained it for 100 seconds—a new record!



Before you leave this page . . .

1. Describe one way in which people or technology are helping to reduce the harmful impacts of energy transformation.

CONCEPT 3

We learn from our mistakes and each other to reduce harmful impacts of energy transformation.

As humans, we make mistakes. However, we can also learn from those mistakes. Often, we do this by changing how we look at things. Our perspectives shift because we have more experience or knowledge. We can look back and consider our actions from a different viewpoint. On the other hand, we may have the benefit of seeing an issue through another's eyes, and now see things differently as a result. In either case, we gain insight into what we might do differently. This is the first step in fixing what went wrong. This applies to reducing harmful impacts of technology-related energy transformation, just as it does to any other problem.

Reducing Harmful Effects by Learning from Past Mistakes

In 2011, a powerful earthquake struck the coast of Japan. Soon after, a 15 m tsunami flooded the Fukushima Daiichi nuclear power plant. A power outage occurred, but the backup generators needed to run the reactors' cooling mechanisms were also disabled by the flooding. The cores of the three operating reactors melted, and several explosions occurred. The fuel tubes melted and burned. Radioactive materials escaped into the atmosphere and water. Almost 20 000 people were exposed to radiation during cleanup efforts (**Figure 3.40**).

How can we learn from accidents such as the one at Fukushima to prevent them from happening again? After the accident, Japan stopped any plans to expand its nuclear program and launched detailed safety inspections of all its existing nuclear reactors. The reactors involved in

the accident were older models with one important design flaw: they still required emergency power to cool during a power failure. Some nations, such as Germany and Italy, have turned away from nuclear power since the accident. In terms of learning from the past, these nations have taken steps to ensure that such an accident does not occur in their country. Other nations, including Japan, still support the transformation of nuclear to electrical energy. However, they are focusing on developing safer technology.

Figure 3.40 The accident at the Fukushima Daiichi nuclear power plant gives us a chance to learn from the past. **Investigating:** Find out how the meltdown at the Fukushima Daiichi nuclear plant was handled and what lessons have been learned since.



Activity

Share Your Knowledge

Create a documentary about how society can learn from a harmful impact of an energy transformation that interests or concerns you. What information you include is up to you. The following questions can help you plan your documentary.

- What is the energy transformation?
- How does society benefit from this transformation?
- How is society harmed by this transformation?
- How can we learn from the harmful impact(s)?

Managing Canada's Used Nuclear Fuel

At this time, there are four nuclear power plants in Canada—three in Ontario and one in Nova Scotia—with 18 reactors in operation. Local communities, and Canada as a whole, face the issue of how to safely manage and store nuclear wastes. The Nuclear Waste Management Organization (NWMO) is responsible for dealing with nuclear wastes in accordance with Canadian laws. This includes finding and making decisions about an appropriate location to bury the material so that it cannot harm life or the land.

The work of the NWMO is an example of collaboration between Indigenous knowledge and Western science. To ensure Indigenous perspectives are included in the selection of a safe storage site, the NWMO established an Indigenous Knowledge Policy. This policy was developed through a Council of Elders and Youth, including First Nations, Inuit, and Métis representatives. The program for identifying a site is ongoing, with a decision expected in 2023 or later.



Extending the Connections

First Peoples Perspectives on Nuclear Waste Management

Indigenous peoples have a variety of opinions about the presence of nuclear waste in their territories. Some support working with industry to find the safest storage method. Others believe no nuclear material should ever enter their lands. Investigate the issue by finding out the NWMO's progress in reaching a solution, how involved Indigenous advisors are in the process, and arguments by people for and against the plan.



Before you leave this page . . .

1. Is it possible to reduce harmful effects of energy transformation by learning from past mistakes? Justify your response using examples from this Concept.

Make a Difference

Should we use nuclear power?

Two main concerns influence public opinion about nuclear power: danger of nuclear accidents and risks related to nuclear wastes. In 2010, the Organization for Economic Co-operation and Development (OECD) conducted a study of international public opinion. Results showed that even though previous nuclear accidents had a negative impact on public opinion about safety, support for nuclear energy had been rising strongly over the past 10 to 20 years in the United States, Western Europe, and Japan. Despite concerns about its waste, nuclear energy was viewed as a dependable energy source. It also reduced a country's dependence on energy imports and contributed less to climate change.

The 2011 accident at Fukushima, Japan, changed things. A study of 23 countries was completed a year after the accident. It showed that attitudes varied by country and were strongly tied to culture and emotions. For example, core support remained strong in the U.S. and Britain. However, Germany

and Italy shifted to an anti-nuclear stance. More consistent was the public's perception. The public now thought that the release of information about major nuclear accidents was too slow and contradictory. As well, there was little trust that institutions can prevent or manage accidents. Even so, the public still views nuclear power as more dependable than renewable sources, such as solar and wind.

Evaluate and Communicate

1. Choose a position for or against the use of nuclear power. Work in a group to prepare for a debate. Consider issues such as safety, costs, waste, reliability, and climate change. If there is time, carry out the debate.
2. What role should public opinion, including your own, play in determining whether nuclear power is part of Canada's energy plan? Write a short (250-word) opinion paper expressing your view.



Check Your Understanding of Topic 3.4

QP Questioning and Predicting **PC** Planning and Conducting **PA** Processing and Analyzing **E** Evaluating
AI Applying and Innovating **C** Communicating

Understanding Key Ideas

1. Use a graphic organizer to outline three positive effects and three negative effects that energy transformation have had on humans. **PA AI C**
2. The photo shows smog over the Fraser Valley.



- a) How is the generation of smog linked to energy transformation by humans?
 - b) How is smog harmful to human health? **PA AI**
3. Draw a comic strip that shows one way that you personally could reduce a negative impact of technology-related energy transformation. **PA AI C**
 4. Why is learning from the past important to our energy future? **PA AI E**
 5. a) Describe how cement and fuel cell technology can both reduce the effects of climate change.
b) Which technology do you think will have a greater effect on the atmospheric concentration of carbon dioxide? Why? **PA AI E C**

Connecting Ideas

6. Consider the tokamak fusion reactor.
 - a) Describe the energy transformation that takes place in the reactor.
 - b) Describe the system that is involved in this energy transformation.
 - c) Describe the surroundings of the system.
 - d) How does the reactor allow fusion to occur in a controlled way?
 - e) Identify the different forms of energy that play role in your answer to part d). **PA AI**

Making New Connections

7. During NASA's space shuttle missions, fuel cells were used to provide electrical energy for the shuttles. The photo shows one of the three fuel cells onboard a shuttle being removed. Use your knowledge of fuel cells to evaluate why a fuel cell was a good choice for this application. **PA AI E**



Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing Information
- Evaluating
- Communicating

What You Need

- access to information resources (for example: online, print, interviews)

Energy Transformation Debate

Do technology-based energy transformations benefit us more than they harm us? Or is the reverse true? The answer to this question is “It’s debatable.” In this investigation, you will join the debate yourself.

Question

Do technology-based energy transformations benefit us more than they harm us? Or is the reverse true?

Procedure

1. Your teacher will assign you to a debate team. With your team, develop a research plan to find support for your position. Consider the following.
 - What energy transformations will you research? For example, in addition to those that involve fuel combustion in vehicles and generating electrical energy, you might consider those involved in agriculture and food industries, product manufacturing, medicine, crime detection, etc. As well, you might consider the benefits and drawbacks of using radioactive isotopes to diagnose and treat diseases, or the risks and costs to food producers and consumers of food irradiation.
 - What questions do you want to answer?
 - What resources will you use?
 - What role will each group member play in completing the research?
 - How will you record your findings and cite your resources?
2. As part of your research plan, you will need to assess your resources for bias and reliability. For each resource you use, consider the following questions.
 - What political positions are represented? Are they obvious or hidden?
 - What special interests might be involved in the issue? Who stands to gain from presenting a particular viewpoint? Who is paying for the message?
 - What sources are used as evidence in the communication? How reliable are they?

- Are facts and statistics cited in the presentation? Are they reliable? Are citations provided so you can check the sources?
 - Is the story one-sided, or are alternate views presented?
 - Are arguments based on facts and logic, or are they emotional appeals? Does it matter?
3. Complete your research and use your findings to prepare your argument for the debate.
 4. Carry out your debate.

Evaluate

1. As a class, decide which team won the debate and why. Do you agree with the class assessment? Justify your response.
2. Why was it important that you assessed your resources for bias and reliability in this Investigation? Explain your answer, making reference to specific resources you used as part of your research.

